

Java Backend Development Notes

A small test document for Spring AI PDF ingestion and RAG

1. Java and Object-Oriented Programming

Java is a high-level, object-oriented programming language. A class is a blueprint used to create objects. An object is an instance of a class and contains state and behavior. The four commonly discussed principles of object-oriented programming are encapsulation, inheritance, polymorphism, and abstraction.

2. Java Collections

The Java Collections Framework provides interfaces and classes for storing and manipulating groups of objects. List represents an ordered collection and allows duplicate elements. ArrayList is a resizable-array implementation of List. LinkedList is based on a doubly linked list. Set represents a collection that does not allow duplicate elements. Map stores data as key-value pairs.

3. HashMap

HashMap is a commonly used implementation of the Map interface. It stores key-value associations and uses hashing to support efficient lookup. A key's hash code helps determine a bucket. During lookup, the hash code helps locate the bucket and equals is used to check whether keys match. Correct implementations of hashCode and equals are important when objects are used as keys.

4. Exception Handling

Java provides exception-handling mechanisms such as try, catch, finally, throw, and throws. Checked exceptions are checked by the compiler, while unchecked exceptions generally occur at runtime. Exception handling allows an application to respond to unexpected situations in a controlled way.

5. Spring Boot

Spring Boot simplifies the development of Java applications by providing auto-configuration, dependency management, embedded servers, and production-ready features. Common annotations include @SpringBootApplication, @RestController, @Service, and @Repository. Dependency injection allows Spring to provide the objects that a class depends on.

6. REST APIs

REST APIs allow applications to communicate using HTTP. GET is commonly used to retrieve resources, POST to create resources, PUT or PATCH to update resources, and DELETE to remove resources. Common HTTP status codes include 200 OK, 201 Created, 400 Bad Request, 401 Unauthorized, and 403 Forbidden.

7. Spring Data JPA and SQL

Spring Data JPA provides repository abstractions for working with relational databases. An entity commonly represents persistent data and can use annotations such as `@Entity` and `@Id`. SQL is used to retrieve and manipulate relational data. `SELECT` retrieves records, `INSERT` adds records, `UPDATE` changes records, and `DELETE` removes records.

8. Interview Revision

Important Java backend topics include OOP, collections, HashMap internals, hashCode and equals, exception handling, multithreading, Spring Boot, dependency injection, REST APIs, JPA, and SQL. Understanding why a technology works is more useful than memorizing definitions alone.

End of test document.